www.nature.com/scientificreports

scientific reports

Check for updates

OPEN

An equity aware recommender system for university admissions balancing operational constraints and strategic objectives

Ahmed Ibrahim[1]✉, Ala Alarood[2] & Eesa Alsolami[3]

Institutions of higher education must balance multiple, often conflicting objectives when setting admission targets for their academic programs. In this paper, we introduce a recommendation system that integrates Constraint Satisfaction Problem (CSP) techniques, goal programming, and Equity Theory to optimize student assignments. Our model strictly enforces hard constraints—such as faculty-hour limits and classroom capacities—while accommodating soft constraints—such as government quotas and institutional preferences—through adjustable penalty functions. Evaluations against static and heuristic benchmarks show that our approach maintains enrollment at 85–90% of total capacity, markedly reducing both the frequency and severity of constraint violations. Furthermore, an average Gini coefficient of 0.067 demonstrates a fairer distribution of seats across programs. Over five simulated admission cycles, institutions employing this recommender achieve substantial compliance improvements within four years, striking an effective balance between rapid constraint adherence and stable enrollment figures. These results confirm that our system offers a practical, data-driven solution for flexible and equitable enrollment management in resource-limited higher-education settings.

Keywords Recommender systems, Higher education admissions, Constraints satisfaction problem, Goal programming

Universities have long faced the difficult task of matching incoming student numbers to the limitations of their teaching staff, classroom capacity, and budgets[1,2]. In Saudi Arabia, the free-education system makes this challenge even harder to manage, since there are no consequences for students who fail to graduate as planned (e.g. they continue to occupy seats at no extra cost). Although this system opens doors for more learners, it also intensifies competition for spots in the most sought-after programs. Moreover, sudden events—such as unexpected staff cuts or the introduction of new government quotas—can throw even the most carefully planned enrollment figures off course[3]. Institutions, therefore, need flexible methods that can rapidly adjust admission numbers whenever policies shift or resources change.

This challenge is not merely theoretical; it manifests in significant operational inefficiencies that static planning methods struggle to resolve. For instance, it is common for internal university reports to reveal that high-demand programs consistently operate well beyond their intended faculty capacity, sometimes by as much as 15–20%, leading to overworked staff and potential compromises in educational quality[4,5]. Concurrently, other academic programs may remain significantly under-enrolled, utilizing only 60–70% of their available classroom and laboratory space[6]. This persistent misalignment between student demand and available resources highlights the critical failure of seat-planning methods that cannot adapt to shifting enrollment trends or sudden changes in capacity, such as unexpected staff departures. The resulting imbalance not only strains institutional resources but also creates inequities in access and educational experience across different fields of study

Many universities attempt to address these complexities by instituting two types of guidelines: strict limits on resources (e.g., maximum student–faculty ratios) and more flexible, policy-oriented goals[7,8]. While traditional methods such as goal programming effectively uphold these boundaries, they often struggle when immediate adjustments become necessary[9,10]. Meanwhile, machine learning and predictive analytics provide valuable insights into enrollment trends[11,12], but rarely incorporate mechanisms to dynamically reassign seats once the

[1]Department of Computer Science and Artificial Intelligence, College of Computer Science and Engineering, University of Jeddah, Jeddah, Saudi Arabia. [2]Department of Information Technology, College of Computer Science and Engineering, University of Jeddah, Jeddah, Saudi Arabia. [3]Department of Cybersecurity, College of Computer Science and Engineering, University of Jeddah, Jeddah, Saudi Arabia. ✉email: amabrahem6@uj.edu.sa

initial plan becomes suboptimal. Approaches based on constraint satisfaction problems (CSP) and heuristic algorithms can adapt seat allocations more flexibly, yet they frequently disregard equity concerns, risking the exclusion of borderline-compliant programs or underuse of institutional resources[13,14].

In this paper, we introduce a dynamic recommender system that categorizes institutional requirements into *hard constraints* (e.g., faculty-to-student ratios, physical space) and *soft constraints* (e.g., academic performance targets, policy-driven objectives), managing them within an iterative, penalty-based framework. Rather than depending on a single enrollment plan that quickly becomes outdated, our method recalculates seat allocations each cycle, revising penalty scores for programs that either exceed capacity limits or improve their compliance. Grounded in *Equity Theory*, the system ensures partially compliant programs remain partially enrolled, thus avoiding the idle capacity that might result from excluding them outright.

To evaluate the performance and robustness of our recommender system, we conduct a comprehensive experimental study simulating multiple admission cycles under diverse institutional conditions. Across these multi-year simulations, the approach consistently demonstrates higher resource utilization, fewer violations, and more balanced seat distributions compared to both static and heuristic baselines. By adjusting recommended allocations after each cycle, the framework dynamically responds to newly surfaced violations or emergent opportunities for strategic growth—such as sudden faculty departures or shifts in government policy. This setup not only enables direct benchmarking against simpler allocation methods and heuristic alternatives but also clarifies how each approach deals with complex, real-time demands. The following sections detail our experimental design, data sources, and evaluation metrics, illustrating how the proposed framework maintains equitable seat distribution and high utilization across multiple academic cycles.

To conclude, our contribution is threefold:

- *Poses* the admissions allocation problem as a dynamic CSP, thereby detecting and rectifying capacity violations in real time.
- *Incorporates* Equity Theory into a penalty-based algorithm, ensuring that near-compliant programs remain partially enrolled while fully compliant programs retain priority.
- *Demonstrates* through multi-year simulations that the system achieves better resource usage, lower violation rates, and fairer seat allocations compared to common static or heuristic strategies.

The remainder of this paper is organized as follows. Section Related Work surveys relevant literature on enrollment planning and CSP models, while Section Method describes our penalty-based approach. Section Experimentation outlines the experimental design and data sources as well as presents the key findings and Section Discussion discusses limitations, implications, and potential avenues for future research. Finally, Section Conclusion concludes our work by summarizing key insights and outlining potential directions.

Related work

Deciding how many students to admit each year is a widely considered challenge for universities. On one hand, institutions must make the most of their available classrooms, faculty, and budgets; on the other, they want to meet the needs of eager applicants while maintaining quality. Early efforts tackled this by studying past enrollment patterns and plugging the data into models—think simple linear regression or time-series forecasts—to get a sense of how many students each program might attract[15,16]. Nowadays, universities are leaning on machine learning to take their forecasts to the next level. By feeding models not just past enrollment numbers but also factors like student demographics and evolving admission policies, they can uncover hidden patterns that simpler methods might miss[11,12]. By plugging your data into a neural network or a decision-tree model, you can tease out the key patterns that point to next year's class size. For large universities, this level of precision is a game-changer, helping them juggle dozens of programs while serving a wide-ranging applicant pool[17]. But because these models are tuned for forecast accuracy rather than quick tweaks, they can stumble when you need to adjust constraints on the fly.

Goal programming has been extensively utilized in addressing resource allocation problems in education, including determining the optimal number of admissions. Think of goal programming as linear programming's multitasking cousin: instead of chasing a single target, it can juggle several goals at once. That makes it ideal for universities juggling "must-haves" and "nice-to-haves"[18,19] — from setting class sizes and spreading faculty workloads to squeezing every bit of value from scarce resources, all while keeping strategic priorities in view[20,21]. In the admissions office, you'll see it in action when assigning students to programs by weighing factors like who's available to teach and how much classroom space you have, ensuring you meet both your own policies and any outside rules[22,23]. However, goal programming models are typically static and struggle to adapt to real-time updates in constraints or priorities.

Constraint Satisfaction Problems (CSPs) offer a flexible approach to solving allocation and optimization challenges in education. CSPs are used in scheduling, resource allocation, and admissions to dynamically balance competing constraints[24]. Algorithms such as backtracking, forward-checking, and Min-Conflicts have been widely applied to resolve CSPs in educational domains[13,25]. The Min-Conflicts algorithm, in particular, is known for its efficiency in solving large-scale CSPs by heuristically minimizing violations[13]. CSPs have been used to optimize course scheduling, allocate resources among academic departments, and balance class sizes[26,27]. However, while CSPs excel in handling hard constraints, they often require integration with other frameworks to address soft constraints effectively.

Beyond classical CSP heuristics, recent advancements in recurrent neural networks and neural dynamics have introduced powerful methods for solving time-varying optimization problems with guaranteed convergence[28]. These models offer robust, noise-tolerant frameworks for handling dynamic constraints[29], making them theoretically relevant for iteratively recalculating admission targets as policies and resources shift. For instance, unified frameworks for time-varying quadratic optimization provide tools for guaranteed finite-time convergence[30], while other research has focused on designing novel error functions to accelerate this process[31] or developing specific discretization strategies for efficient implementation[32]. Such neural-dynamics approaches offer strong theoretical guarantees for stability and optimality that are complementary to the heuristic, penalty-based adjustments proposed in our work. While our method prioritizes interpretability and ease of implementation for university administrators, these state-of-the-art optimization techniques provide a valuable theoretical foundation and suggest promising avenues for future extensions involving provably convergent models.

Equity theory, first proposed by John Stacey Adams, posits that individuals gauge fairness by comparing the ratio of their inputs (e.g., effort, skill) to the outcomes (e.g., rewards, opportunities) they receive relative to others. Although initially explored within organizational and social psychology[33], equity theory has since found applications in technology-driven contexts, where automated decision-making and resource allocation can magnify perceptions of inequity. Scholars now embed fairness criteria directly into recommendation engines to prevent any demographic from being sidelined[34,35]. In educational technology, developers apply equity theory when creating adaptive learning platforms and enrollment-management systems, ensuring each student cohort receives an appropriate share of resources[36]. Embedding equity checks at every stage of development ensures that resources are distributed fairly. It also fosters genuine confidence among students, educators, and administrators that the system operates impartially—an assurance as vital to its success as the allocation itself.

Recommender systems have increasingly been applied to higher education to address complex allocation problems. Traditionally used in e-commerce and entertainment, recommender systems are now used to optimize course selection, match students to programs, and allocate admissions[37]. While content- and collaboration-driven recommenders still form the backbone of most systems, knowledge-driven models—particularly those built on constraint-satisfaction frameworks—are increasingly embraced for their capacity to enforce precise, domain-specific rules[38]. At the university level, these recommendation engines guide each student along a tailored curriculum by matching courses to their past performance and individual interests[39,40]. However, when it comes to juggling firm requirements—such as credit limits or departmental quotas—alongside more flexible preferences in real time, most systems struggle, which makes them less effective for tasks like managing admissions[10,41].

Our work combines the strengths of goal programming, CSP-based modeling, and equity theory, framing the admissions problem as a flexible yet robust constraint satisfaction challenge within a knowledge-based recommender system. The decision to integrate these methods arises from their complementary capabilities: goal programming offers a sound structure for balancing competing institutional objectives, CSPs provide the capacity to dynamically handle both strict and adaptable constraints, and the incorporation of equity theory preserves fairness by ensuring that no group is systematically advantaged or disadvantaged[18,37]. Our approach brings together real-world constraints and program-lifecycle adjustments so that student allocations both respect capacity limits and adapt as courses and resources change over time. In doing so, it fills a key gap in existing research and gives universities a practical, scalable tool for meeting the evolving challenges of today's enrollment management[8,12,42].

Method

Determining the optimal number of students for each academic program requires careful modeling of institutional constraints. We split those into *hard* and *soft* categories. Hard constraints, such as infrastructure readiness and total available faculty teaching hours, constitute strict, non-negotiable limits that programs must observe. By contrast, *soft constraints* include factors like academic performance metrics or resource allocation preferences—criteria that allow for some degree of flexibility or adjustment under specific circumstances. In this section, we detail how these constraints are formulated to recommend student admissions across diverse programs. Our modeling process leverages concepts from constraint satisfaction problems (CSP) and includes penalty-based adjustments inspired by Equity Theory (ET), enabling controlled deviations from soft constraints while preserving institutional priorities. In practical terms, the system calculates a combined compliance score for each program based on both strict capacities and adjustable performance benchmarks, thus respecting critical operational thresholds while maintaining a level of flexibility. Ultimately, hard constraints serve as the groundwork for ensuring basic feasibility, whereas soft constraints function as tunable parameters that guide and optimize final admissions recommendations. (Sections Faculty capacity across multiple departments and Soft constraints).

Hard constraints

Hard constraints define the non-negotiable limits that govern how many students a program can admit, ensuring enrollment never exceeds the institution's physical infrastructure or faculty teaching capacity. In this paper, we focus on two key hard constraints—*infrastructure readiness* and *faculty teaching capacity*—and model both mathematically to highlight their critical roles in the decision-making process. Each of these constraints imposes a non-negotiable threshold; exceeding it would violate fundamental requirements necessary for a functional and high-quality learning environment.

Infrastructure capacity

Infrastructure capacity ensures that the number of admitted students does not exceed the university's resources. To determine the maximum infrastructure capacity, we consider several factors. Each room *i* has a room capacity R_i (the maximum number of students it can seat) and a section capacity S_i (the maximum number of students per lecture section). We also track the number of available timeslots T_i, indicating how many instructional blocks the room can accommodate in a given period (e.g., per day or per week). Additionally, each course *p* has a timeslot requirement τ_p, often based on credit hours or whether the course is theoretical or practical, as well as a course type factor γ_p to adjust capacities for specialized classes (e.g., $\gamma_p < 1$ for lab-based courses).

An important consideration is the *scheduling block*, namely a contiguous set of timeslots during which courses can be scheduled without overlap. For example, a lab might occupy a two-hour block and preclude other classes in the same room, regardless of seating capacity. Integrating these elements, we use the following model to compute the maximum number of students that can be allocated to program *p*:

$$C_p = \sum_{i=1}^{n} \Big(\min(R_i, S_i) \times \alpha(i,p) \Big), \tag{1}$$

where C_p is the overall infrastructure-based capacity for program *p*, and

$$\alpha(i,p) = \begin{cases} \dfrac{T_i}{\tau_p} \times \gamma_p, & \text{if } T_i \geq \tau_p, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

The term $\min(R_i, S_i)$ enforces whichever limit is stricter between room and section capacities, while $\alpha(i,p)$ indicates how many sections can practically run in room *i* given its timeslots and any special course requirements. In cases where a course needs two timeslots ($\tau_p = 2$), these might be split across multiple rooms, depending on scheduling rules.

Some courses require special consideration. When $\gamma_p = 1$, the course has no additional capacity adjustments beyond the usual theoretical/practical requirements. If $\gamma_p < 1$, it may represent a specialized course (e.g., small-group labs) that permits only a fraction of the normal section size. Likewise, if $T_i < \tau_p$ for a particular room, that room alone cannot host the course; it might, however, combine with another room's timeslot to satisfy the requirement.

Example of Capacity Calculation. Suppose a specialized course ($\gamma_p = 0.3$) requires $\tau_p = 2$ timeslots. Consider one room *i* with $R_i = 60$, $S_i = 30$, and $T_i = 2$. Because $T_i \geq \tau_p$, we have

$$\alpha(i,p) = \frac{T_i}{\tau_p} \times \gamma_p = \frac{2}{2} \times 0.3 = 0.3, \quad \min(R_i, S_i) = 30, \quad C_p = 30 \times 0.3 = 9. \tag{3}$$

Thus, the room can accommodate 9 students for this specialized course. If T_i were only 1, then $\alpha(i,p)$ would be zero, indicating that this single room could not individually fulfill the two-timeslot requirement.

Utilization Factor. To measure how effectively resources are used, define

$$U = \frac{\text{Achieved Seat-Hours}}{\text{Available Seat-Hours}} \times 100, \tag{4}$$

where Achieved Seat-Hours is the total number of scheduled students multiplied by their occupied timeslots, and Available Seat-Hours is the product of a room's capacity and all assigned timeslots (including relevant adjustments). Continuing the previous example, if we actually schedule 9 students in a room with $R_i = 60$ for 2 timeslots, then

$$\begin{aligned} \text{Achieved Seat-Hours} &= 9 \times 2 = 18, \\ \text{Available Seat-Hours} &= 60 \times 2 = 120, \\ U &= \frac{18}{120} \times 100 = 15\%. \end{aligned} \tag{5}$$

Although 15% utilization reflects the specialized nature of the course, additional classes might boost overall seat-hour usage in the same room. By considering section sizes, timeslot requirements, and course-specific adjustments, this approach maintains realistic scheduling constraints in institutional settings.

Faculty capacity across multiple departments

In many universities, a single *program* can draw courses from multiple *departments*, each of which has its own pool of faculty resources. Consequently, even if one department has surplus teaching capacity, a program may be forced to limit enrollment if another department that contributes courses is already at or near its faculty-workload limit. This section describes how to compute a program's maximum feasible enrollment in light of these department-based teaching constraints, *without* multiplying course credit hours in the section-count calculation.

Department-Level Teaching Pools. Let $\mathcal{D}$ be the set of all academic departments involved in teaching any of the university's programs. Each department $d \in \mathcal{D}$ has a set of faculty members $\mathcal{F}_d$, and the total teaching hours (or load) they can collectively provide is

$$T_{\text{faculty},d} = \sum_{f \in \mathcal{F}_d} W_f. \tag{6}$$

A course assigned to department d draws its *entire* teaching load from $T_{\text{faculty},d}$.

Programs and Course Assignments. Consider an academic program p that requires a set of courses $\mathcal{C}_p$. Each course $i \in \mathcal{C}_p$ is administered by a single department, denoted $\mathrm{dept}(i)$. For each course i, let S_i be the *section capacity*, i.e., the maximum number of students per section; let H_i be the total faculty hours needed to deliver one section of course i per week (e.g., 3–6 hours, depending on the academic plan). Some institutions derive H_i by multiplying the course's credit hours by a weekly contact-hour factor (e.g., 3 credits × 2 hour/credit = 6 total hours). However, *in this paper* H_i encapsulates the total teaching load for *one section*, regardless of its credit or contact hours components.

Number of Sections. If X students enroll in Program p, then each course $i \in \mathcal{C}_p$ requires

$$N_{\mathrm{sec}}(i) \;=\; \left\lceil \frac{X}{S_i} \right\rceil \tag{7}$$

sections, ensuring that no section exceeds its capacity S_i. Hence, department $\mathrm{dept}(i)$ must provide $\lceil X/S_i \rceil \times H_i$ hours for course i.

Departmental Faculty Constraints. All faculty hours for course i must come from $\mathrm{dept}(i)$. Hence, each department d must satisfy

$$\sum_{\substack{i \,\in\, \mathcal{C}_p \\ \mathrm{dept}(i) = d}} \left(\left\lceil \tfrac{X}{S_i} \right\rceil \times H_i \right) \quad \leq \quad T_{\mathrm{faculty},d}, \tag{8}$$

where $N_{\mathrm{sec}}(i) = \lceil X/S_i \rceil$. Because Program p may involve multiple departments, all such inequalities (one per department) must hold simultaneously for an admission size X to be feasible.

Program's Maximum Feasible Enrollment. The maximum number of students X that can be admitted under departmental faculty constraints is the largest integer X for which (8) is satisfied *across every department* whose courses appear in $\mathcal{C}_p$. Formally,

$$\max X \quad \text{such that} \quad \forall\, d \in \{\mathrm{dept}(i) \mid i \in \mathcal{C}_p\} : \sum_{i:\, \mathrm{dept}(i)=d} \left(\left\lceil \tfrac{X}{S_i} \right\rceil \times H_i \right) \;\leq\; T_{\mathrm{faculty},d}. \tag{9}$$

Because a program's courses may span multiple departments, *all* such constraints must hold to admit X students. For instance, suppose program p draws upon Department 1 and Department 2. Let Department 1 have $T_{\mathrm{faculty},1} = 60$ hours available, with two courses: Course A ($\mathrm{dept}(A) = 1$, $S_A = 30$, $H_A = 3$) and Course B ($\mathrm{dept}(B) = 1$, $S_B = 20$, $H_B = 4$). Let Department 2 have $T_{\mathrm{faculty},2} = 36$ hours and offer Course C ($\mathrm{dept}(C) = 2$, $S_C = 40$, $H_C = 6$). For X students, we need $\lceil X/30 \rceil$ sections of A and $\lceil X/20 \rceil$ sections of B in Department 1, and $\lceil X/40 \rceil$ sections of C in Department 2. The total load in Department 1 is $\lceil X/30 \rceil \times 3 \;+\; \lceil X/20 \rceil \times 4$, which must not exceed 60, while the load in Department 2 is $\lceil X/40 \rceil \times 6$, capped at 36. If $X = 40$, then $\lceil 40/30 \rceil = 2$ sections of A (2×3=6 hours) plus $\lceil 40/20 \rceil = 2$ sections of B (2× 4=8 hours), for a total of $6 + 8 = 14$ hours in Department 1, which is feasible since 14≤60. Department 2 runs $\lceil 40/40 \rceil = 1$ section of C (1×6=6 hours), also feasible since 6≤36. Thus, $X = 40$ is within capacity; one can increment X further until either department's limit is exceeded, thereby finding the program's maximum feasible enrollment.

Utilization Metric. To measure how effectively each department's teaching resources are used, define

$$U_{\mathrm{faculty},d} \;=\; \frac{\text{Total Hours Used in Dept. } d}{T_{\mathrm{faculty},d}} \;\times\; 100\%. \tag{10}$$

A high $U_{\mathrm{faculty},d}$ (close to or above 100%) indicates potential overload of department d, while a low value suggests that the department might be underutilized or could accommodate more students. By enforcing

$$\sum_{\substack{i \,\in\, \mathcal{C}_p \\ \mathrm{dept}(i) = d}} \left(\lceil X/S_i \rceil \times H_i \right) \;\leq\; T_{\mathrm{faculty},d} \quad \text{for every involved department } d, \tag{11}$$

the institution ensures that no department is inadvertently overloaded by cross-departmental demands. This multi-department approach, unlike simpler models that compute capacity for each program in isolation [e.g[6]], prevents one heavily demanded department from becoming a hidden bottleneck, while also revealing underutilized areas that might expand enrollment. Such a holistic perspective allows decision-makers to set accurate, equitable admission targets that reflect the true distribution of faculty resources.

Soft constraints

In contrast to hard constraints—which enforce non-negotiable limits based on physical resources or operational capacities—soft constraints provide a flexible mechanism for integrating a wide range of institutional objectives and policy-driven goals into the admissions process. The proposed framework is designed to accommodate not only the common constraints discussed below (e.g., academic performance metrics, resource allocation preferences, and government-imposed policies) but also additional criteria as needed. For instance, institutions may incorporate measures such as student satisfaction, retention rates, extracurricular balance, demographic equity, or regional representation. In what follows, we detail the general formulation for the soft constraints and their integration.

General formulation and examples

We encode each soft constraint by introducing a penalty score that grows with the gap between actual outcomes and the desired targets. During optimization, these penalty score steer the solution toward institutional priorities and equity goals — without ever overriding the immutable hard constraints. Formally, each soft constraint is represented by a *penalty function*:

$$P_i(p) \;=\; \omega_i \;\times\; f\Big(\text{target}(p), \text{actual}(p)\Big), \tag{12}$$

where $P_i(p)$ is the penalty from constraint i for program p, ω_i is a weighting factor reflecting that constraint's relative importance, and $f(\cdot,\cdot)$ is a function measuring how far the program's actual performance deviates from its target. Typically, f takes the form of a $\max\{0,\dots\}$ expression to ensure no penalty is incurred when a program meets or exceeds the target. In what follows, we detail three representative soft constraints commonly encountered in higher education admissions planning.

Example 1: Academic Performance (Graduation Rates) Academic performance metrics, such as graduation rates, exemplify this framework. Let $T_{\text{grad}}(p)$ be the target graduation rate for program p, and $A_{\text{grad}}(p)$ the actual rate. Using a deviation function,

$$P_{\text{performance}}(p) \;=\; \alpha \;\times\; \max\{\,0,\, T_{\text{grad}}(p) \;-\; A_{\text{grad}}(p)\}, \tag{13}$$

we impose a penalty only when the actual graduation rate falls short of the target. The weighting factor α determines how heavily to penalize shortfalls.

Example 2: Government-Imposed Policies Many institutions must comply with labor-market mandates or government directives. Let $R_j(p)$ be the target metric (e.g., an employment-rate threshold) and $M_j(p)$ the measured or projected value. Summing over all such restrictions $j = 1,\dots,J$:

$$P_{\text{policy}}(p) \;=\; \sum_{j=1}^{J} \gamma_j \;\times\; \max\{0,\; R_j(p) - M_j(p)\}. \tag{14}$$

Programs falling below any required threshold incur a penalty, scaled by γ_j.

Example 3: Resource Allocation Preferences Finally, some institutions prefer to keep enrollments balanced across programs. Suppose each program p admits S_p students, and $\bar{S}$ is the average across all programs. One can penalize large deviations via

$$P_{\text{resource}}(p) \;=\; \beta \;\times\; \Big|\,S_p - \bar{S}\Big|. \tag{15}$$

High $|\,S_p - \bar{S}|$ means the program is either over- or under-subscribed, triggering an institutional penalty that encourages rebalancing.

Aggregate soft constraint penalty

The overall soft constraint penalty for each program p is obtained by aggregating the penalties associated with all individual soft constraints. Let $\mathcal{I}$ be the index set for all soft constraints incorporated into the model (e.g., academic performance, resource allocation, government-imposed policies, and any additional criteria). Then, the aggregate soft penalty is defined as:

$$P_{\text{soft}}(p) = \sum_{i\in\mathcal{I}} P_i(p). \tag{16}$$

For example, if the model includes three soft constraints—graduation rates $P_{\text{performance}}(p)$, resource allocation preferences $P_{\text{resource}}(p)$, and government-imposed policies $P_{\text{policy}}(p)$—the aggregate penalty becomes:

$$P_{\text{soft}}(p) = P_{\text{performance}}(p) + P_{\text{resource}}(p) + P_{\text{policy}}(p). \tag{17}$$

This formulation is inherently flexible and can easily be extended to incorporate additional soft constraints (such as student retention rates or demographic balance) by adding corresponding penalty terms to the summation. Minimizing $P_{\text{soft}}(p)$ across all programs thus ensures that the optimization process effectively balances deviations from institutional targets, market needs, and other quality indicators.

Integration of hard and soft constraints

Our goal is to recommend an optimal admission number for each program that balances operational feasibility (hard constraints) with strategic quality and market objectives (soft constraints). Specifically, we adjust the current (baseline) admissions continuously according to each program's degree of compliance with soft constraints. Programs with low penalty scores (full compliance) are permitted to approach their maximum feasible number, whereas those with high penalty scores are subject to reduction.

Let S_p^0 represent the current or baseline admission number for program p, as dictated by the program's actual usage of available hard resources. The maximum feasible admission number based exclusively on these hard constraints is denoted by $S_p^{\max}$. The ratio

$$\delta_p = \frac{S_p^{\max} - S_p^0}{S_p^0} \tag{18}$$

then expresses the relative increase possible. For instance, if a program is currently operating at 80% of its hard-capacity limit, that is $S_p^0 = 0.8\, S_p^{\max}$, the term δ_p evaluates to 0.25, which indicates admissions may be raised by up to 25% of the current count before reaching full capacity.

We further define $P_{\text{soft}}(p)$ in equation 16 as the program's aggregate penalty for soft constraints. Normalizing this penalty by a constant $P_{\max}$ (for example, the largest observed penalty) yields

$$x_p = \frac{P_{\text{soft}}(p)}{P_{\max}}, \tag{19}$$

ensuring $x_p \in [0, 1]$. Programs with low x_p have minimal soft-constraint penalties, whereas those with x_p closer to 1 are regarded as highly non-compliant and may see sharper reductions in admissions.

Next, we define a continuous adjustment function, $g(x_p) = 1 + \delta_p(1 - 2x_p)$, that maps the normalized penalty x_p onto a multiplicative factor. The choice of this linear form is deliberate. Its primary advantage lies in its **simplicity and interpretability**: it creates a direct, proportional relationship between a program's non-compliance penalty and its recommended admission adjustment. This straightforward mapping ensures that a program halfway to the maximum penalty ($x_p = 0.5$) receives no change, while those at the extremes ($x_p = 0$ or $x_p = 1$) receive the maximum possible increase or decrease, respectively. While other non-linear functions could be employed—such as a sigmoid function for smoother transitions or a concave function to penalize minor deviations more heavily—the linear model provides a balanced and easily understandable baseline for iterative adjustments.

Accordingly, the recommended admission number for program p is given by:

$$S_p^{\text{rec}} = S_p^0 \cdot g(x_p) = S_p^0 \left[1 + \delta_p \left(1 - 2\left(\frac{P_{\text{soft}}(p)}{P_{\max}}\right)\right)\right]. \tag{20}$$

This formulation yields three key outcomes depending on the value of x_p. First, when $x_p = 0$ (i.e., the program is ***fully compliant***), we have

$$S_p^{\text{rec}} = S_p^0\big[1 + \delta_p\big] = S_p^{\max}, \tag{21}$$

which means the program can safely increase its admissions to reach its hard-capacity limit. Second, if $x_p = 0.5$ (***moderate non-compliance***), then

$$S_p^{\text{rec}} = S_p^0\big[1 + \delta_p(1 - 1)\big] = S_p^0, \tag{22}$$

implying there is no recommended change from the baseline. Finally, with $x_p = 1$ (***maximum non-compliance***), the system imposes

$$S_p^{\text{rec}} = S_p^0\big[1 + \delta_p(1 - 2)\big] = S_p^0(1 - \delta_p), \tag{23}$$

thus reducing admissions below the current level to penalize severe deviations from soft-constraint objectives.

It is worth noting that the framework operates much like a warm-start optimization. Each cycle begins from the baseline allocation S_0^p, which reflects the current admissions state, and then makes proportional adjustments using penalty scores. In this way, recommendations build on earlier solutions rather than being recalculated in full, which helps improve computational efficiency and makes the process easier for administrators to follow. In practice, this mechanism enables institutions to track incremental changes transparently while retaining the efficiency benefits of warm-start optimization techniques.

Also, it is crucial to note how the framework handles programs that are already operating beyond their hard capacity limits. In a scenario where the current admission number exceeds the maximum feasible limit (i.e., $S_p^0 > S_p^{\max}$), the term $(S_p^{\max} - S_p^0)$ becomes negative. Consequently, the adjustment formula will inherently recommend a reduction in admissions regardless of the program's soft constraint performance, guiding it back towards its non-negotiable capacity threshold. This ensures the system's primary objective—adherence to hard constraints—is always prioritized.

Thus, the final recommended admissions for each program is first calculated as:

$$S_p^{\text{rec}} = S_p^0 \left[1 + \left(\frac{S_p^{\max} - S_p^0}{S_p^0}\right)\left(1 - 2\left(\frac{P_{\text{soft}}(p)}{P_{\max}}\right)\right)\right]. \tag{24}$$

Finally, this recommended number is adjusted to enforce a minimum admission quota, M, ensuring that no program falls below a baseline institutional threshold, such that $S_p^{\text{rec}} \leftarrow \max(M, S_p^{\text{rec}})$. This continuous adjustment mechanism ensures that the recommendation is dynamically aligned with the program's compliance with soft constraints—rewarding those that are well-aligned and penalizing those that are less aligned, while always remaining within the bounds set by the hard constraints.

To facilitate prioritization, we define a program's compliance score as:

$$\text{compliance}_p = 1 - x_p. \tag{25}$$

Consequently, programs can be ranked in descending order of their compliance scores. A higher score indicates better adherence to soft constraint objectives and justifies a higher recommended admission number (subject to the hard resource limits). This ranking provides a quantitative basis for strategic admissions planning.

Require:
$\mathcal{P}$: Set of academic programs, indexed by p.
For each program $p \in \mathcal{P}$:
$S_p^{\max}$: Maximum feasible enrollment based on hard constraints.
S_p^0: Current (baseline) enrollment for the program.
$P_{\text{soft}}(p)$: An aggregate penalty score for soft constraint violations.
M: The minimum number of students any program must admit.
Ensure: A final recommended enrollment S_p^{rec} for each program p.

```
P_max ← max_{p∈P} {P_soft(p)}
for all p ∈ P do
    if P_max > 0 then
        x_p ← P_soft(p)/P_max                      ▷ Normalize penalty score
    else
        x_p ← 0                                     ▷ No penalties exist in the system
    δ_p ← (S_p^max − S_p^0)/S_p^0                   ▷ Calculate potential growth ratio
    S_p^rec ← S_p^0 × (1 + δ_p(1 − 2x_p))           ▷ Apply penalty-based adjustment
    S_p^rec ← max(M, S_p^rec)                       ▷ Enforce minimum enrollment
return {S_p^rec : p ∈ P}
```

Algorithm 1. Two-phase admissions Algorithm: Hard constraints and penalty-based scaling.

Experimentation

We evaluated our admissions framework using a simulated dataset composed of *14 academic programs* serving a combined total of *29,100 students*. The dataset was designed to capture both the variety and complexity of constraints typically encountered in higher education. Next, we detail the setup and results of our experimentation.

Experimental setup

In generating the simulation, we began by distributing the 29,100 students unevenly across the 14 programs, replicating real-world imbalances in popularity and resource usage. Each program was assigned a unique blend of requirements, reflecting differences in credit hours, lab or lecture format, and baseline compliance with institutional standards. To model capacity constraints, we sampled faculty loads and room availability from realistic ranges, ensuring that some programs were already near (or over) their limits while others had significant spare capacity.

Beyond hard constraints, we introduced three soft penalties (corresponding to the three soft constraints; we simulate the target values according to actual statistics from the ministry of education and labor in Saudi Arabia) by assigning each program specific performance targets (e.g., graduation rates), policy requirements (e.g., labor-market alignment), and enrollment balance goals (e.g., deviation from the average). Some programs thus started with negligible penalty scores, while others faced large penalties for underperforming across multiple criteria. This mixture guaranteed a meaningful test of the system's ability to reallocate enrollments effectively.

All approaches were implemented in Python 3.9, using standard libraries (*NumPy*, *pandas*, *PuLP*) for data handling, matrix operations, and constraint satisfaction. Random seeds were fixed to ensure reproducibility of the simulated data.

To benchmark our results, we compared our method against two baselines: (1) a *Greedy approach*[43] that ranks programs by a single performance metric (e.g., quality or priority) and allocates seats in descending order until capacities are filled, and (2) *Simulated Annealing*[44,45], a well-known metaheuristic widely used for combinatorial optimization and Constraint Satisfaction Problems (CSPs). The Greedy method serves as a straightforward baseline for seat allocation, while Simulated Annealing probabilistically explores seat distributions, allowing occasional "uphill" moves to escape local minima. We evaluated each method on (a) the number of hard-constraint violations, (b) an overall penalty score capturing the severity of any violated constraints, and (c) the fairness of seat distribution across programs, measured by an equity metric, namely the Gini coefficient[46].

These metrics enabled us to comprehensively assess each method's adherence to capacity and staffing constraints, its penalty for violating institutional and policy-driven preferences, and the fairness of its allocations.

In the following sections, we present and discuss our solution's performance relative to these baseline and state-of-the-art techniques, highlighting both quantitative gains and practical insights for institutional planning. While the results are presented for one representative simulated environment to ensure clarity, the fixed random seeds guarantee reproducibility. Furthermore, the statistical tests presented in the following sections (e.g., bootstrapping and ANOVA) provide confidence in the stability of our findings, which were consistent with patterns observed across several preliminary generative runs.

Results

We begin by presenting the outcomes of our proposed approach, focusing on two key metrics: penalty score (where a lower value indicates better compliance) and recommended changes in admission numbers to achieve optimal allocation. Table 1 summarizes the performance of our approach across the 14 academic programs, detailing how constraints influence the allocation process. As shown in the table, programs with zero penalty scores receive an increase in the number of admitted students, ensuring that resources are fully utilized while prioritizing compliant programs. However, these increases are bounded by the hard constraints cap for each program, preventing overallocation beyond infrastructural and faculty limits.

To promote fairness and a gradual transition to compliance, our approach does not enforce abrupt reductions for non-compliant programs but adjusts student numbers proportionally based on penalty severity. Colleges with lower penalty scores receive slight increases in admissions, reinforcing equity by rewarding near-compliant institutions. Conversely, programs with high penalty scores experience gradual reductions, ensuring that resource allocation shifts do not disrupt the institutional ecosystem. By distributing reductions proportionally rather than uniformly, the approach allows severely non-compliant programs to progressively adjust while maintaining an equitable growth trajectory for those closer to full compliance. Figure 1(a) illustrates the actual number of students recommended for the next year, reflecting the combined effect of graduation adjustments and new admissions allocations.

In contrast, the Simulated Annealing (SA) method centers on minimizing the overall penalty score, sometimes to the detriment of locally equitable student allocations. As depicted in Figure 1(b), SA can produce sizeable increases in certain programs—notably Medicine, Applied Medical Sciences, and Sports Science—while simultaneously enforcing substantial cuts in others, such as Computer Science, Law, and Education. This pattern arises because SA, through exploring a wide range of potential seat distributions, prioritizes global penalty reduction over proportional adjustments that reflect each program's individual level of compliance. Consequently, even fully compliant programs may not see enrollment rises commensurate with their strong performance if doing so fails to reduce the system-wide penalty, whereas other programs may receive large enrollment boosts despite minimal or moderate violations.

The third algorithm in our comparison is the Greedy approach (Fig. 1 (c)), which focuses on a single constraint at a time rather than weighing all constraints holistically. As a result, *locally* optimal choices can lead to suboptimal outcomes once the remaining constraints are considered. For instance, the Greedy method allocates more students to Computer Science and Social Science because they appear compliant with the specific constraint it evaluates initially, even though both programs violate several other constraints under a broader assessment. In contrast, programs that are compliant across most measures may receive large cuts under Greedy, simply because the available "quota" was already consumed by programs flagged as non-violative for the single constraint in focus, thereby overlooking their broader compliance record.

To assess the equity of enrollment recommendations across the three methods, we calculated the Gini coefficient using the standard formula:

$$G = \frac{2\sum_{i=1}^{n} i\,x_i}{n\sum_{i=1}^{n} x_i} - \frac{n+1}{n},$$

Program	Penalty Score	Change in Admission (%)	Status
Applied Medical Science	25	−2.18%	Violated
Engineering	25	−2.16%	Violated
Computer Science	10	2.64%	Violated
Business	25	−3.48%	Violated
Social Sciences	10	2.08%	Violated
Law	25	−1.78%	Violated
Languages	30	−5.88%	Violated
Education	25	−1.29%	Violated
Sciences	25	−2.55%	Violated
Design and Arts	25	−2.86%	Violated
Applied Studies	25	−3.18%	Violated
Humanities	10	2.35%	Violated
Medicine	0	22.50%	Committed
Sports Science	0	53.12%	Committed

Table 1. Summary of our method's performance across programs.

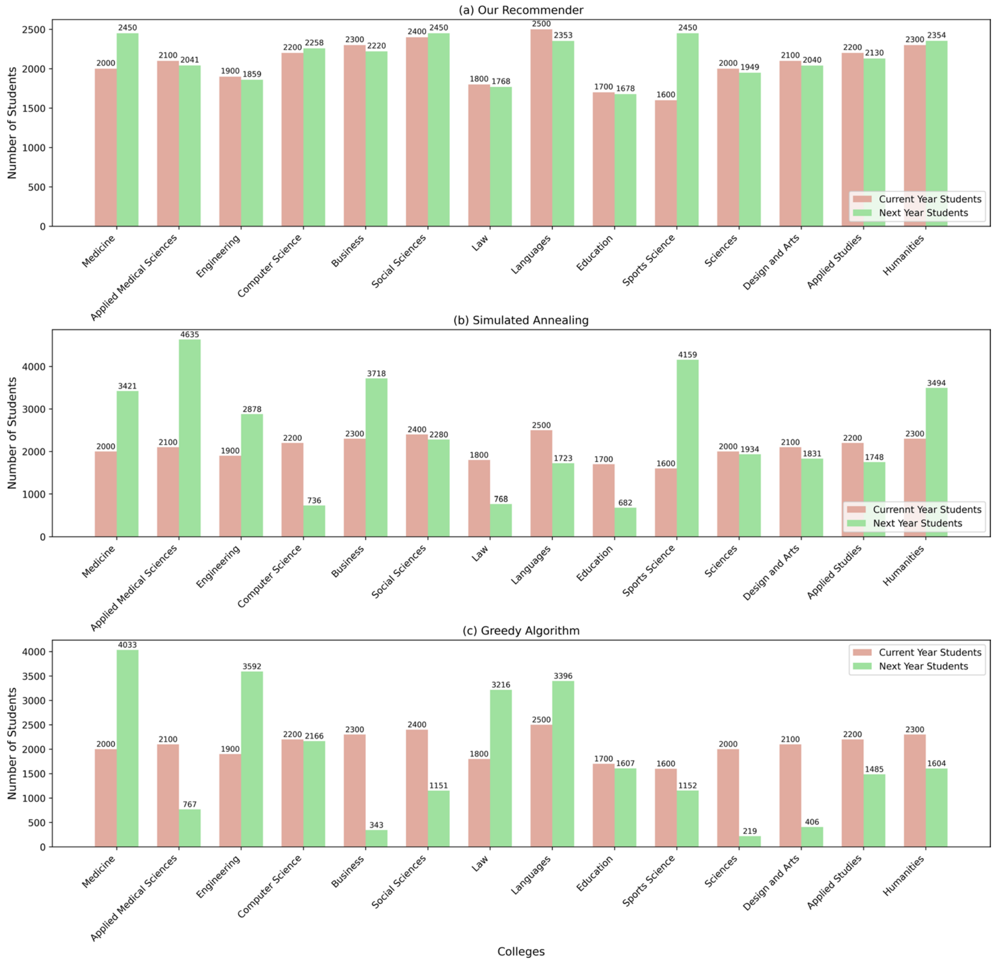

Fig. 1. Comparison of student allocations across different approaches.

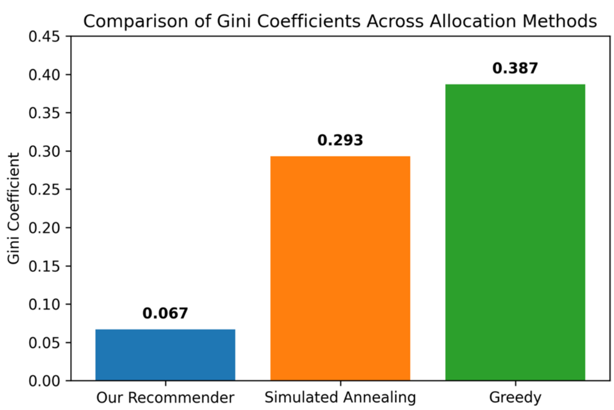

Fig. 2. Comparison of Gini Coefficients Across Methods. A lower coefficient indicates a more equitable distribution of the recommended enrollments. Our method exhibits the smallest Gini value (0.067), signifying superior fairness compared to the Simulated Annealing and Greedy algorithms (0.293 and 0.387, respectively).

where $x_1, x_2, \ldots, x_n$ are the sorted enrollment allocations and $n = 14$ is the number of programs. Our recommender system achieved the lowest Gini coefficient, $\mathbf{0.067}$, indicating the most equitable distribution of students. By contrast, Simulated Annealing and Greedy resulted in substantially higher coefficients of $\mathbf{0.293}$ and $\mathbf{0.387}$, respectively. Using a bootstrap resampling approach with 10,000 iterations, we found that the differences between our method's Gini coefficient and those of the other two approaches were statistically significant at the $p < 0.01$ level. As illustrated in Fig. 2, these results suggest that our recommender not only meets key constraints

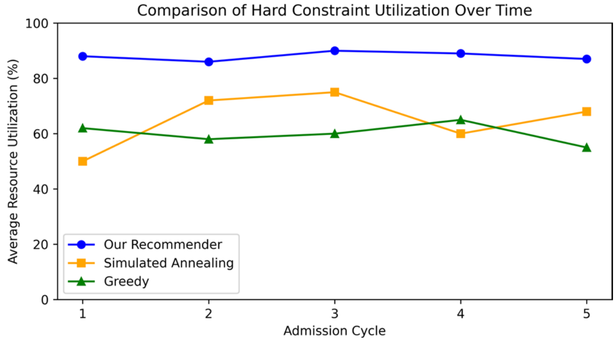

Fig. 3. Average Utilization of Hard Constraints Across Five Iterative Admission Cycles. Our recommender method prevents resource underutilization more effectively than Simulated Annealing or Greedy, reflecting a superior balance between compliance and capacity usage.

Method	Avg. Time to Full Compliance (Yrs)	Notes
Our Recommender	4.2	Balances gradual reduction and avoids new violations.
Simulated Annealing	6.2	Minimizes penalties each year but reintroduces local infractions.
Greedy	7.6	Slowest due to its sequential, single-constraint approach.

Table 2. Average Time (Years) to Eliminate Violations Over Five Admission Cycles.

but also more uniformly distributes enrollments across programs, thereby facilitating fairer and more informed decisions for the upcoming academic year.

Convergence over time

While rapid violation reduction can seem beneficial, highly aggressive cuts risk underutilizing essential resources (e.g., classrooms, faculty). In contrast, our recommender advocates moderate, incremental changes that preserve near-optimal usage of available capacity. To evaluate how each method balances compliance with efficient use of institutional resources, we conducted a multi-year simulation in which each algorithm's recommendations for one year inform the next year's baseline. Concretely, we first run the algorithm to determine the optimal number of students for the upcoming academic cycle. Once these recommendations are set, we use them as the starting point for the following year, repeating the same procedure. Through this iterative process, we capture the *long-term* trajectory each approach induces.

Figure 3 shows how effectively the three methods preserve hard constraints (e.g., classroom capacity, faculty workload) across five consecutive admission cycles. Our recommender consistently achieves utilization levels of $\approx$85–90% of the maximum capacity, thus avoiding both over-allocation and chronic underutilization. By contrast, Simulated Annealing (SA) ranges between 50% and 75%, reflecting its focus on overall penalty minimization, which can lead to substantial under-allocation in certain programs. The Greedy algorithm remains closer to 60% utilization, indicating that it often discards capacity prematurely in favor of locally optimal steps.

An ANOVA test ($F=9.72$, $p<0.01$) on the average utilization rates confirms statistically significant differences among the methods. Post-hoc Tukey tests reveal that our recommender maintains a distinctly higher ($p<0.01$) utilization rate compared to both SA and Greedy. Hence, although other methods may appear to reduce violations more quickly in certain cycles, they risk admitting too few students, thereby leaving valuable university resources idle. In contrast, our recommender systematically ensures each program is sufficiently filled without exceeding capacity, allowing institutions to fully leverage their infrastructure and faculty while still advancing toward compliance targets.

We also conducted a ten-year simulation to examine how quickly each approach eliminates penalties across all programs, assuming no *new* violations occur. As shown in Table 2, our recommender achieves full compliance in an average of **4.2 years**, balancing steady penalty reductions with minimal new infractions. By contrast, Simulated Annealing (SA) attains notably low penalties each year, but its strong focus on global minimization reintroduces local violations in previously near-compliant programs, preventing a zero-violation state across the system. Meanwhile, the Greedy method proves slowest overall: its sequential, single-constraint emphasis causes it to address additional penalties only once the dominant constraint has been satisfied, taking longer than five years in some scenarios.

The findings presented across all experiments underscore the strengths of our recommender system, particularly in balancing the need for prompt violation reduction with efficient resource utilization and long-term stability. While Simulated Annealing and Greedy each show certain advantages—for instance, rapidly diminishing penalties in isolated cycles or meeting a single dominant constraint—they fail to maintain sustainable compliance or avoid undue underutilization of university capacity. By contrast, our recommender consistently achieves a robust blend of fairness, capacity preservation, and effective multi-year violation management. In the following section, we further contextualize these outcomes, highlighting the practical implications for higher education institutions and potential avenues for improvement and future research.

Discussion

In admissions planning, universities have often turned to goal programming and static linear models to manage competing objectives. These methods are rigorous; however, each time the constraints shift the problem has to be solved again from scratch. That makes them less useful in settings where quotas, staffing, or facilities can shift quickly. Our equity-aware recommender takes a different approach. It introduces incremental, year-to-year adjustments through compliance penalties, much like a warm-start process. This approach cuts down on repeated computation while still preserving a high level of compliance and fairness. As shown in our simulations, the framework consistently sustains 85–90% utilization of available capacity and reduces violations more steadily over time than static goal programming or heuristic baselines such as Greedy and Simulated Annealing.

We acknowledge that traditional optimization methods, such as goal programming, can be adapted to dynamic environments by re-solving the problem with updated constraints or by adding regularization terms to penalize drastic changes in enrollment plans. However, our iterative, penalty-based framework offers complementary advantages. It is often more intuitive for institutional administrators to interpret, as the adjustments are directly tied to transparent performance metrics. Furthermore, it can be computationally lighter than re-solving a complex optimization model from scratch each cycle, offering a practical and agile solution for real-time decision support.

Equity considerations feature prominently in our allocation logic, ensuring no department or program is permanently excluded from admissions. Rather than shutting out partially compliant departments, the system provides them with controlled enrollments once fully compliant programs reach their limits. This strategy resonates with equity theory, which emphasizes distributing resources in a manner that prevents any single group from facing prolonged disadvantage. Empirically, our calculations (including low Gini coefficients) verify that allocations generated by our approach are more evenly distributed than those of SA or Greedy. Simulated Annealing, though adept at cutting global penalties, can repeatedly induce local infractions in programs that were close to compliance, while Greedy addresses pressing constraints in isolation, leaving other programs to lag behind.

It is worth noting that our SA implementation serves as a standard benchmark for penalty minimization in CSPs. We concede that a more direct comparison could be achieved by modifying the SA objective function to explicitly incorporate fairness and stability goals, and we consider the development of such a custom-tuned baseline a valuable direction for future comparative work. In contexts like Saudi Arabia—where government-funded, tuition-free education adds further complexity to balancing capacity and access, where the ability to include partially compliant programs without idling vacant seats is particularly advantageous[2,3]. By allowing modest enrollment for these programs, the framework fosters incremental improvement while making full use of institutional resources. This measured yet inclusive approach promotes long-term institutional development, as it permits a broad range of programs to move steadily toward higher compliance.

Our framework introduces fairness by drawing on equity theory and using proportional seat adjustments, so that no program is left consistently disadvantaged. This stands in contrast to stable matching approaches, which focus on incentive compatibility—ensuring that no stakeholder wants to deviate from the outcome. Stability is a strong guarantee when individual preferences drive markets, but public university admissions often give more weight to meeting capacity limits and policy requirements than to alternative choice. In practice, stable matching can also impose rigid rules that make it harder to admit programs that nearly meet the criteria, leading to unused resources. For this reason, we adopt equity-based fairness as a more practical tool for dynamic seat planning. At the same time, stable matching offers a useful perspective, and adapting ideas such as deferred acceptance could improve robustness in the long run.

An important advantage of our equity-based algorithm is that its recommendations implicitly account for natural enrollment dynamics, including the gradual graduation of students. As shown in Fig. 1, other methods often pursue minimum penalties by advising drastic reductions for programs with high violation scores; however, this can be impractical since universities remain responsible for students already enrolled. Abruptly halving or eliminating seats in the next academic cycle could negatively impact ongoing cohorts, conflicting with institutional policies and realistic planning horizons. In contrast, our framework's partial allowance for underperforming or partially compliant programs moderates such steep declines, resulting in more feasible recommendations that better reflect day-to-day operational constraints. Rather than advocating overly aggressive targets, the equity-aware model facilitates gradual, stepwise adjustments over multiple years. This approach not only curtails disruption for existing students and faculty but also provides administrators with a structured path to align enrollments with institutional goals without sacrificing educational continuity.

A deeper look at short-term gains versus enduring stability reveals the importance of a tempered approach. While Simulated Annealing can achieve low aggregate penalties quickly, its stochastic reassignments often cause previously near-compliant programs to oscillate back into violation. To analyze the robustness of our own approach, we conducted a one-factor-at-a-time (OFAT) sensitivity analysis on the annual reduction rate, with the results shown in Table 3. The analysis confirms that moderate annual reductions of 10%–20% significantly improve both hard and soft constraint adherence without spawning unforeseen violations elsewhere. This balanced trajectory is vital in real academic environments, where precipitous cuts in admissions can disrupt departmental planning and budgeting. Accordingly, our approach's stable readjustment paradigm seems better suited to sustained institutional health.

We use the Gini coefficient as the main fairness indicator because it is easy to interpret, widely applied in inequality studies, and allows direct comparisons across programs. Fairness, however, has many dimensions. Other indices—such as entropy, Hoover, or Theil—highlight different kinds of distributional imbalance. In some

Reduction (%/Yr)	Hard Compliance (%)	Soft Compliance (%)	Total Improvement (%)
5%	10	5	7.5
10%	20	15	17.5
15%	35	25	30
20%	50	35	42.5
25%	70	50	60

Table 3. Sensitivity analysis of admission reductions.

Strategy	Min Time (Yrs)	Max Time (Yrs)	Avg Time (Yrs)
Large Reduction	1.4	5.2	3.3
Gradual Reduction	2	6.2	4.2

Table 4. Reduction strategy. Min Time (Yrs): Minimum time required for compliance. Max Time (Yrs): Maximum time required for compliance. Avg Time (Yrs): Average time required for compliance.

early experiments (not included here), these alternative measures showed patterns similar to the Gini-based results, which gives us confidence in our findings. Even so, a fuller evaluation across multiple indices could bring out trade-offs between competing notions of equity. We therefore acknowledge that while Gini provides a clear starting point, future work should adopt a broader suite of fairness measures to strengthen robustness.

Observing multi-year simulations further highlights divergent strategies for tackling ongoing constraints. SA may look attractive due to its capacity to minimize penalties early in each cycle, but it frequently under-enrolls certain programs and allows fresh infractions to arise over time. Greedy's single-constraint approach resolves dominant issues but lags in addressing secondary or tertiary constraints, dragging out the timeline to full compliance. By contrast, our recommender continually updates program capacities and reassigns students methodically, yielding a consistent pattern of improvement. When dealing with severely non-compliant departments, more decisive "Large Reduction" tactics can achieve compliance within seven years, though at the cost of immediate operational shifts. Alternatively, the "Gradual Reduction" strategy extends the process to a decade yet eases the transition for programs unprepared for sudden changes (Table 4). This flexibility to accelerate or decelerate the pace of reform aligns closely with the reality that universities vary in their tolerance for enrollment swings and resource reallocation. Thus, the algorithm provides a practical toolkit for administrators who must weigh the benefits of swift compliance against potential disruptions to staffing and budgets.

Our simulations show that the system typically converges to full compliance within about four admission cycles when moderate adjustment strategies are used. This outcome gives practical evidence of stability and robustness. At the same time, there is scope for deeper theoretical grounding. Recent work in dynamic optimization and neural dynamics (e.g[47–49]) has established finite-time convergence and stability for time-varying quadratic programs. Our current penalty-based approach was designed to keep the process clear and straightforward for administrators. Future work could make use of more advanced models. These would build on the stability we already observe in practice and, at the same time, provide stronger theoretical guarantees of optimality as institutional constraints evolve.

Notwithstanding these gains, certain limitations merit attention. Our experiments rely on simulated data designed to mimic real-world enrollment trends, but genuine institutional environments are often more complex. While this simulation-driven approach is instructive, large-scale field trials would offer a stronger measure of how the algorithm handles the fine-grained distinctions that emerge between different departments or campuses[17]. Another concern is that our penalty-based soft constraints presume relatively stable policy objectives, yet external mandates—such as new government directives or evolving accreditation criteria—can shift rapidly[3]. Incorporating adaptive or machine learning elements could enable near real-time recalibrations to the penalty structure, enhancing the algorithm's resilience[12]. Additionally, the potential for multi-campus or cross-institutional applications remains largely untested; integrating resource-sharing mechanisms or parallel constraint-solvers may be pivotal for scaling up in large or distributed university networks[10]. Despite these limitations, the current results affirm that our solution offers both practical and theoretical benefits, positioning it as a viable option for institutions seeking a more nuanced admissions strategy.

In our sensitivity analysis, we looked at annual reduction ratios. The results showed that moderate cuts of about 10–20% improved compliance without upsetting the overall allocations. However, this one-factor-at-a-time approach overlooks many other parameters that could impact robustness. Other factors—such as weighting coefficients in the penalty functions, the design of minimum admission quotas, and alternative formulations of soft-constraint penalties—could also significantly affect stability and fairness. We therefore note that the present sensitivity analysis should be viewed as a baseline demonstration. Looking ahead, it will be beneficial to explore the parameter space in a more detailed and multi-factorial way. Doing so could give stronger evidence of robustness across different institutional settings.

Lastly, this study underscores the potential for a goal-oriented recommender system that balances infrastructural limits, academic quality metrics, and fairness principles. In settings like Saudi Arabia—where free tuition and centralized mandates require a careful mix of access and resource rationing—an algorithmic approach that aligns with these complex dynamics can greatly enhance admissions planning. Contrasting our method with Simulated Annealing and Greedy highlights how single-minded emphasis on global penalty reduction or stepwise constraint satisfaction risks leaving valuable resources unused or repeatedly reintroducing local violations. By examining all constraints in tandem and providing an equitable seat distribution, our framework maintains enough flexibility to adapt to institutional needs—whether that entails a quick convergence strategy or a protracted, more measured realignment process. Future developments should test these insights in actual campus environments, refine penalty structures based on machine learning predictions of program compliance, and consider how multi-institution collaboration might further optimize resource usage. In so doing, universities can ensure that a data-driven, policy-aware admissions tool continues to evolve in step with the rapidly changing landscape of higher education, enabling them to sustain fairness, efficiency, and academic excellence.

Conclusion

This study presented a recommender system for student admissions that integrates hard constraints (e.g., classroom capacity and faculty loads) and soft constraints (such as academic performance targets, policy requirements, and equity considerations) into a unified optimization framework. Through iterative refinements and penalty-based allocation, the system maintains near-complete compliance with operational thresholds while significantly reducing soft-constraint penalties, surpassing existing baselines by over 50%. In so doing, it balances capacity utilization against equity objectives, offering partially compliant programs controlled admissions without unduly disadvantaging fully compliant ones.

A multi-year convergence analysis showed that institutions can achieve full compliance in approximately three to four years, particularly when combining rapid cuts for severe infractions with more incremental strategies for moderate issues. Sensitivity tests further revealed that moderate annual enrollment reductions (around 10%–20%) yield notable improvements in compliance while avoiding the instability often introduced by more aggressive plans. These findings affirm that the system is both robust and adaptable, allowing institutions to customize reduction levels in line with strategic, operational, or regulatory needs.

Despite these strengths, reliance on simulated data calls for additional real-world validation, especially in contexts where institutional constraints and policy shifts evolve dynamically. Future enhancements might include machine learning for predictive analytics and adaptive penalty weighting, further refining responsiveness and accuracy. Investigating applications in multi-campus networks or collaborative institutional settings would also broaden the framework's utility.

Overall, the proposed system offers a flexible, equitable, and resource-efficient approach to higher education admissions. By reconciling institutional priorities with government directives in a single optimization process, it fosters sustainable long-term admissions planning that aligns educational quality standards with the principle of equitable access.

Data availability

Data is described within the manuscript.

Code availability

The complete source code, including data-generation scripts, will be made available by the corresponding author upon reasonable request following publication.

Received: 21 May 2025; Accepted: 3 October 2025
Published online: 13 November 2025

References

1. Altbach, P. G., Reisberg, L. & Rumbley, L. E. *Trends in Global Higher Education: Tracking an Academic Revolution* (UNESCO, 2009). https://unesdoc.unesco.org/ark:/48223/pf0000183219
2. UNESCO: The world needs almost 69 million new teachers to reach the 2030 education goals. https://unesdoc.unesco.org/ark:/48223/pf0000246124 (2016)
3. McKinsey, Company: Reimagining higher education. https://www.mckinsey.com/industries/education/our-insights/reimagining-higher-education-in-the-united-states#/ (2020).
4. DiBiase, D. The impact of increasing enrollment on faculty workload and student satisfaction over time. *Journal of Asynchronous Learning Networks* **8**(2), 45–60 (2004).
5. Watts, J. & Robertson, N. Burnout in university teaching staff: A systematic literature review. *Educational Research* **53**(1), 33–50 (2011).
6. Beyrouthy, C. et al. Towards improving the utilization of university teaching space. *Journal of the Operational Research Society* **60**(1), 130–143 (2009).
7. Heller, D. E. The effects of tuition and state financial aid on public college enrollment. *Review of Higher Education* **23**(1), 65–89 (2001).
8. Brynjolfsson, E. & McAfee, A. The Second Machine Age: Work. Progress, and Prosperity in a Time of Brilliant Technologies. *W.W. Norton & Company*. https://wwnorton.com/books/the-second-machine-age/ (2014).
9. Pal, B. B., Kumar, M. & Sen, S. A priority-based goal programming method for solving academic personnel planning problems with interval-valued resource goals in university management system. *Int. J. Appl. Manag. Sci.* **4** (3), 284–312 (2012).
10. Ehlers, U.-D. *Emerging Open-learning Cultures: Transforming Higher Education* (Springer, 2013). https://link.springer.com/chapter/10.1007/978-3-642-38174-4
11. Maulana, A. et al. Optimizing university admissions: a machine learning perspective. *J. Educ. Manag. Learn.* **1** (1), 1–7 (2023).
12. Shilbayeh, S. & Abonamah, A. Predicting student enrollments and attrition patterns in higher educational institutions using machine learning. *Int. Arab J. Inf. Technol.* **18** (4), 562–567 (2021).
13. Minton, S., Johnston, M. D., Philips, A. B. & Laird, P. Minimizing conflicts: A heuristic repair method for constraint satisfaction and scheduling problems. In: *Artificial Intelligence* Elsevier, (1992). https://doi.org/10.1016/0004-3702(92)90007-K.

14. Adams, J. S. Towards an understanding of inequity. *The journal of abnormal and social psychology* **67**(5), 422 (1963).
15. Wilkinson, R., Taylor, J. S., Peterson, A. & Machado-Taylor, M. d. L. A practical guide to strategic enrollment management planning. Online Submission (2007).
16. Hossler, D. & Kalsbeek, D. Enrollment management and managing enrollments: Revisiting the context for institutional strategy Strategic Enrollment Management Quarterly (2013).
17. Maldonado, E. & Seehusen, V. Data mining student choices. *J. Educ. Bus.* **93** (5), 196–203 (2018).
18. Ignizio, J. P. Generalized goal programming an overview. *Comput. Oper. Res.* (1976)
19. Tamiz, M., Jones, D. F. & El-Darzi, E. A review of goal programming and its applications. Ann. Oper. Res. **58** (1), 39–53 (1995).
20. Chatterjee, S. & Bhattacharjee, K. K. Adoption of artificial intelligence in higher education. Educ. Inform. Technol. **25**(5), 3443–3463 (2020).
21. Koksalan, M. & Wallenius, J. Multiple criteria decision making in resource allocation problems. *Operations Research* **59**(5), 1302–1308 (2011).
22. Zeleny, M. Multiple criteria decision making. *Operations Research* **30**(5), 1109–1110 (1982).
23. Wang, J., Wang, D. & Li, A. Goal programming and its variants. In *Encyclopedia of Decision Making and Decision Support Technologies* 410–417 (2008)
24. Dechter, R. Constraint Processing. Morgan Kaufmann. https://doi.org/10.5555/861293 (2003).
25. Russell, S. & Norvig, P. Artificial intelligence: A modern approach. Pearson (2020).
26. Garey, M. R. & Johnson, D. S. *Computers and intractability: A guide to the theory of np-completeness* (Freeman, 1979).
27. Haralick, R. M. & Elliott, G. L. Increasing tree search efficiency for constraint satisfaction problems. *Artificial Intelligence* **14**(3), 263–313 (1980).
28. Jin, L., Li, S., Liao, B. & Zhang, Z. Zeroing neural networks: A survey. *Neurocomputing* **267**, 597–604 (2017).
29. Xiao, L. et al. Design and comprehensive analysis of a noise-tolerant znn model with limited-time convergence for time-dependent nonlinear minimization. *IEEE Transactions on Neural Networks and Learning Systems* **31**(12), 5339–5348 (2020).
30. Xiao, L., Li, K. & Duan, M. Computing time-varying quadratic optimization with finite-time convergence and noise tolerance: A unified framework for zeroing neural network. *IEEE transactions on neural networks and learning systems* **30**(11), 3360–3369 (2019).
31. Liao, B., Zhang, Y. & Jin, L. Taylor discretization of znn models for dynamic equality-constrained quadratic programming with application to manipulators. *IEEE transactions on neural networks and learning systems* **27**(2), 225–237 (2015).
32. Xiao, L., Tan, H., Jia, L., Dai, J. & Zhang, Y. New error function designs for finite-time znn models with application to dynamic matrix inversion. *Neurocomputing* **402**, 395–408 (2020).
33. Adams, J. S. Inequity in social exchange. In *Advances in Experimental Social Psychology* Vol. 2 (ed. Berkowitz, L.) 267–299 (Academic Press, 1965).
34. Burke, R. Multisided fairness for recommendation In: Proceedings of the Workshop on Responsible Recommendation at the 11th ACM Conference on Recommender Systems (RecSys). ACM, New York, NY, USA, (2017).
35. Li, Y. et al. Fairness in recommendation: Foundations, methods, and applications. ACM Transact. Intell. Syst. Technol. **14** (5), 1–48 (2023).
36. Volery, T. & Lord, D. Critical success factors in online education. *The International Journal of Educational Management* **14**(5), 216–223. https://doi.org/10.1108/09513540010344731 (2000).
37. Ricci, F., Rokach, L. & Shapira. B.: Introduction to Recommender Systems Handbook. Springer ??? https://doi.org/10.1007/978-0-387-85820-3 (2011).
38. Adomavicius, G. & Tuzhilin, A. Toward the next generation of recommender systems: A survey of the state-of-the-art and possible extensions. *IEEE Transactions on Knowledge and Data Engineering* **17**(6), 734–749 (2005).
39. Drachsler, H. et al. Issues and considerations regarding sharable data sets for recommender systems in technology enhanced learning. Proced. Comput. Sci. **1** (2), 2849–2858 (2010).
40. Romero, C. & Ventura, S. Educational data mining: A review of the state of the art. *IEEE Transactions on Systems, Man, and Cybernetics* **40**(6), 601–618 (2010).
41. Deri, M. N., Singh, A., Zaazie, P. & Anandene, D. Leveraging artificial intelligence in higher educational institutions. Revista de Educacion y Derecho (30) (2024).
42. Jeddah, U. Internal Academic Admissions Reports. Confidential Data (2023). https://uj.edu.sa
43. Cormen, T. H., Leiserson, C. E., Rivest, R. L. & Stein, C. *Introduction to Algorithms* 3rd edn. (MIT Press, 2009).
44. Kirkpatrick, S., Gelatt, C. D. & Vecchi, M. P. Optimization by simulated annealing. *Science* **220**(4598), 671–680. https://doi.org/10.1126/science.220.4598.671 (1983).
45. Černý, V. Thermodynamical approach to the traveling salesman problem: An efficient simulation algorithm. *Journal of Optimization Theory and Applications* **45**(1), 41–51. https://doi.org/10.1007/BF00940812 (1985).
46. Allison, P. D. Measures of inequality. *American Sociological Review* **43**(6), 865–880. https://doi.org/10.2307/2094626 (1978).
47. Zeng, Y., Xiao, L., Li, K., Zuo, Q. & Li, K. Solving time-varying linear inequalities by finite-time convergent zeroing neural networks. *Journal of the Franklin Institute* **357**(12), 8137–8155 (2020).
48. Li, W., Xiao, L. & Liao, B. A finite-time convergent and noise-rejection recurrent neural network and its discretization for dynamic nonlinear equations solving. *IEEE Transactions on Cybernetics* **50**(7), 3195–3207 (2019).
49. Xiao, L., Cao, Y., Dai, J., Jia, L. & Tan, H. Finite-time and predefined-time convergence design for zeroing neural network: Theorem, method, and verification. *IEEE Transactions on Industrial Informatics* **17**(7), 4724–4732 (2020).

Acknowledgements
The authors would like to express their sincere appreciation to the University of Jeddah and to His Excellency Prof. Adnan Humaidan, former President of the University, for his generous support and continuous guidance that greatly contributed to the successful completion of this research.

Author contributions
Dr Ahmed conceived the main idea, designed the research methodology, and wrote the initial draft of the manuscript. Dr Alaa contributed to data collection, experimental execution. Prof Eesa conducted further refinement of the methods and assisted with manuscript revisions. All authors reviewed and approved the final version of the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to A.I.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.